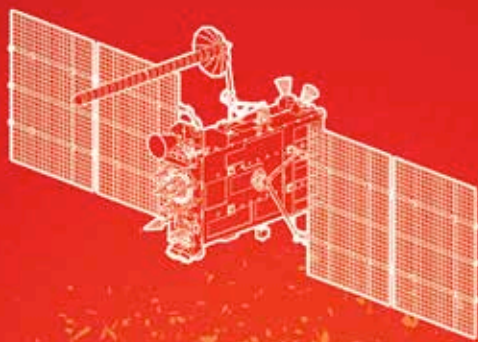




SPACE

WAR IN THE FINAL FRONTIER



With the launch of the Sputnik in 1957, the US became concerned about potential nukes in orbit, and developed anti-satellite (ASAT) technologies that could neutralise targets in orbit. Russia followed suit, and both powers tested their capabilities with launches in orbit, although only on their own satellites. Both the powers realised that assets in orbit were a means of de-escalation and stabilisation of conflict, and were not unduly concerned about the militarisation of space, with the “spy” satellites in orbit only providing strategic inputs. All of that changed with the Gulf War, when the assets in space were used to provide active intelligence to the boots on the ground and the drones in the sky, at times providing near real time support to the armed forces. The effectiveness with which the US used its space assets in the Gulf War led the Chinese and Russian militaries to develop their ASAT capabilities, with a flash point occurring on 11 January 2007, when the Chinese destroyed one of their own defunct

meteorological satellites in orbit using an ASAT missile with a kinetic kill warhead. The satellite disintegrated, producing 3,000 trackable pieces of debris in orbit, out of which about 2,800 continue to circle the Earth till this day. These pieces of debris remain a hazard to the assets of other countries in orbit, including the International Space Station. China was heavily admonished by the international community for the act. Since then, China, the US and Russia have all demonstrated their ASAT capabilities on their own satellites, but none have produced debris that persists in orbit. There has been no deliberate act on enemy satellites so far. India is the fourth country to have demonstrated its ASAT capabilities (after the US, Russia and China), with a modified Prithvi missile taking down the Microsat-R, which was an “Earth observation” satellite deployed into orbit specifically and nefariously to act as

target practice for the ASAT test, known as Mission Shakti. Although the disintegration created debris, India, unlike China was not reprimanded by the international community for the test, because all the debris produced was at a very low altitude and harmlessly reentered the atmosphere of the Earth within a period of a few weeks.

There are a number of ways to neutralize the satellites in orbit. A ground station attack just takes out the communication capabilities on Earth, without harming the satellite in orbit. A direct ascent ASAT is typically a ballistic missile that is modified to reach the satellite in orbit, like India did with its Mission Shakti. In a co-orbital ASAT strike, the weaponised payload is first deployed into the same orbit as the satellite, with the capabilities of striking its target on command. Non kinetic space weapons in development include high power lasers, microwave weapons and EMP weapons that can neutralize satellites in orbit or ground stations without physically making contact with them. Then there are the electronic and cyber ASAT measures, which are differentiated. The former involves jamming or spoofing the communications between the satellites and the ground station, while the latter involves intercepting, monitoring or corrupting the data itself, or seizure of control of the spacecraft. China has been the one of shadiest of all state actors when it comes to ASAT tests, with a number of suspicious maneuvers near the assets of other governments, small periods of hostile takeover of satellites, and interception of the data.



The PIRPL satellite uses its infrared sensor to detect ballistic missiles from orbit

Credit: Space Development Agency